

Where Solution Begins.

AI-Droponics: An IOT-based Hydroponics Monitoring and Automation System with Integrated AI Crop Disease Detection for Gulayan at Bulaklakan Integrated National Urban Farm

By

Concepcion, Justine Altaire B.

De Juan, Axel P.

Lingatong, Nazh Manuel V.

Odon, Marc Joseph V.

Salvador, Gerard Jose N.

Adviser
Maria Carolina Bautista

Subject Teacher
Arnel Balasta

Year 2026-2027

TABLE OF CONTENTS

Chapter 1: Startup Branding..... 1
1.1 Company Name..... 1
1.2 Company Logo..... 1
1.3 Tagline..... 3
1.4 Vision..... 3
1.5 Mission..... 4
1.6 Core Values..... 4
Chapter 2: Project Overview..... 5
2.1 Introduction..... 5
2.2 Problem Statement..... 5
2.3 Proposed Solution..... 6
Chapter 3: Project Overview..... 8
3.1 Innovation Statement..... 8
3.2 Design Thinking Summary..... 8
3.3 Proposed Innovative Features..... 11
3.4 Innovation Value..... 12
Chapter 4: Startup Opportunity Analysis..... 14
4.1 Startup Readiness Assessment..... 14
4.2 Market and Competitor Analysis..... 24
4.3 Technology Trends..... 27
4.4 Unique Selling Proposition (USP)..... 28
Chapter 5: Customer Validation..... 30
5.1 Customer Persona..... 30
5.2 Interview Results..... 31
5.3 Validation Findings..... 32
5.4 Recommended Improvements..... 33

Chapter 1: Startup Branding

1.1 Company Name

Our startup is officially named **Source** because it represents how our team approaches innovation. Each member contributes a unique technical specialization, and by combining these strengths, we are able to develop practical and effective solutions through collaboration. Rather than naming the company after a specific product or technology, we chose a name that reflects our problem-solving process and allows the brand to grow beyond a single innovation.

The name also aligns with our long-term vision. **Source** is the company behind AI-Droponics, our IoT-based hydroponics monitoring and AI crop disease detection system for urban farmers, while also serving as the foundation for future technology-driven products and services. As the origin of our solutions, the name positions the company as a trusted source of innovation that delivers efficient, reliable, and scalable technologies. Its simplicity, memorability, and versatility make it well-suited for a startup with ambitions beyond a single product.

1.2 Company Logo



Figure 1: Company Logo

The Source logo was designed to represent the company's identity as a technology-driven startup that transforms ideas, expertise, and emerging technologies into practical solutions. The overall concept reflects the belief that effective innovation begins by bringing together different perspectives, skills, and technologies to address real-world challenges. Each element of the logo was intentionally designed to communicate Source's identity as a collaborative, forward-thinking, and reliable technology company capable of developing solutions that evolve with the needs of its users and communities.

Meaning of the Logo

The logo represents Source as an origin point for innovative solutions. Its central **"S"** represents the company name, while the overall design conveys connectivity, movement, and continuous innovation. The combination of the central symbol and surrounding form reflects Source's role in connecting ideas, expertise, and technology to create practical solutions.

Color Psychology

The logo uses a **gradient of deep navy to bright cyan**, complemented by a **white background**.

- **Deep Navy** represents trust, reliability, professionalism, and stability, reflecting the qualities Source aims to provide through its solutions.
- **Cyan Blue** represents innovation, intelligence, technology, and forward movement, aligning with the company's focus on AI, IoT, and emerging technologies.
- **White** represents clarity, simplicity, and openness, allowing the logo to remain clean, modern, and recognizable.

The transition from navy to cyan also represents **progress and technological growth**, reinforcing Source's identity as an evolving and forward-thinking company.

References

Adobe. (n.d.). *Color symbolism and meaning*. https://www.adobe.com/uk/creativecloud/design/discover/color-symbolism.ht**ml**

Adobe. (n.d.). *Color theory*. https://www.adobe.com/uk/creativecloud/design/discover/color-theory.html

Canva. (n.d.). *Electric blue color meaning*. https://www.canva.com/colors/color-meanings/electric-blue/

Kliever, J. (n.d.). *Design for engagement*. Canva. https://www.canva.com/learn/design-for-engagement/

Font

The wordmark **"SOURCE"** uses a **bold, geometric sans-serif typeface** with uppercase lettering. This font style was chosen for its clean and modern appearance, communicating **precision, confidence, professionalism, and technological sophistication**. Its simple geometric structure also ensures readability across digital and physical applications, making it suitable for a technology startup designed to grow across different products and industries.

Symbolism

The primary symbol is a stylized "**S**" formed by two interlocking angular shapes. The interlocking forms represent collaboration, connectivity, and the integration of different technologies, reflecting Source's approach of combining expertise and systems to develop solutions.

Surrounding the S is an open circular ring. Its incomplete form represents continuous growth, adaptability, and openness to new ideas, while its circular movement reflects the continuous flow of information within Source's technology-driven solutions.

Together, the S, interlocking shapes, and open ring communicate Source's core identity: connecting expertise and technology to continuously develop intelligent, reliable, and innovative solutions.

1.3 Tagline

Where Solution Begins.

Our tagline reflects the meaning of our company name, **Source**, by positioning the company as the starting point where innovative solutions are created. It reinforces our identity as a team that combines diverse technical expertise to transform ideas into practical, technology-driven solutions.

The tagline also represents our commitment to being a trusted partner for our clients and users. Whether through AI-Droponics or future innovations, Source aims to

be the place people turn to for reliable, efficient, and impactful solutions that address real-world challenges.

1.4 Vision

To become a trusted technology company recognized for building high-quality, intelligent, and sustainable products that solve real-world challenges. Beginning with smart urban farming, Source envisions a future where accessible, reliable, and innovative technologies help farmers, industries, and communities become more productive, resource-efficient, and resilient, improving quality of life.

1.5 Mission

We develop accessible, technology-driven solutions that address real-world challenges through innovation, artificial intelligence, and the Internet of Things. Beginning with AI-Droponics, we empower urban farmers to improve productivity, optimize resources, and adopt sustainable farming practices while continuously creating intelligent and reliable solutions that serve the evolving needs of industries and communities.

1.6 Core Values

Innovation

We continuously explore emerging technologies to develop practical, forward-thinking solutions that address real-world challenges.

Collaboration

We believe that combining diverse skills and perspectives leads to stronger ideas, better decisions, and more effective solutions.

Reliability

We are committed to delivering dependable, high-quality technologies that users can trust in their daily operations.

Sustainability

We create solutions that promote efficiency, responsible resource management, and long-term environmental and social impact.

Growth

We embrace continuous learning, adaptability, and improvement to ensure that our company, our products, and the communities we serve continue to evolve.

Chapter 2: Project Overview

2.1 Introduction

Source is a technology-driven team dedicated to solving real-world agricultural challenges through smart, accessible innovation. In line with this mission, Source identified a critical gap in urban hydroponic farming, where growing conditions are tracked manually, disease detection depends entirely on a farmer's visual inspection, and corrective action from nutrient dosing to treatment application happens reactively rather than in real time.

Even farmers with years of hands-on experience have no automated system to catch early warning signs before disease spreads through a shared water supply. To address this gap, Source developed AI-Droponics, an IoT-based hydroponics monitoring and automation platform paired with AI-powered crop disease detection, giving urban farmers a more reliable and less labor-intensive way to run their operations.

2.2 Problem Statement

Urban farmers practicing hydroponics lack any integrated system for environmental monitoring, disease detection, or automated response, and instead rely on manual, visual, and reactive methods. This creates three core problems:

- **No early disease detection.** Hydroponic crops are vulnerable to root and bacterial diseases (such as Pythium Root Rot and Bacterial Soft Rot) that spread quickly through a shared, recirculating water supply once introduced. Without a reliable way to catch these early, farmers often resort to pesticide use as a default response rather than a targeted intervention.
- **No continuous environmental monitoring.** Manual tracking of pH levels, temperature, and humidity is time-consuming and prone to human error. Without continuous monitoring, farmers often catch fluctuations only after the crop has already been stressed or damaged.

- **Inefficient, manual corrective action.** Disease detection depends entirely on visual inspection by the farmer, which is inconsistent, delayed, and difficult to scale, especially for small teams managing multiple grow beds.

Many farmers, lacking a reliable way to catch these issues early, resort to pesticide use as a default response rather than targeted intervention. Second, manual monitoring of critical growing conditions pH levels, temperature, and humidity is time-consuming and prone to human error.

Without continuous tracking, farmers often catch fluctuations too late, after they've already stressed or damaged the crop. Third, the manual detection of plant diseases depends heavily on visual inspection by the farmer, which is inconsistent, delayed, and difficult to scale, especially for small teams managing multiple grow beds. Together, these problems make hydroponic farming more labor-intensive, less predictable, and harder to sustain profitably at an urban scale.

2.3 Proposed Solution

AI-Droponics is an IoT-based hydroponics monitoring and automation system with AI-powered crop disease detection, purpose-built for urban hydroponic farms such as Gulayan at Bulaklakan. It combines three integrated components:

- **AI-powered crop disease detection.** A Raspberry Pi 5 paired with a camera module captures live leaf images and processes them on-device using a YOLOv8n model, identifying early signs of disease and flagging issues before they spread, a particularly important safeguard given how quickly disease can travel through a shared water-based system.
- **Real-time environmental monitoring.** Connected sensors continuously track pH levels, temperature, and humidity, displaying live readings through a centralized mobile dashboard and removing the need for manual checks.
- **Automated corrective response.** Based on sensor thresholds and AI disease classification, the system automatically triggers nutrient dosing, and water pumping, so corrective action happens without the farmer needing to intervene manually every time.

- **Mobile application.** The companion app functions as a remote control for the entire system, letting farmers set adjustable thresholds for when automation should trigger, watch a live stream of the crop disease detection feed, receive AI-powered treatment suggestions and recommendations, and get real-time push notifications whenever a disease is detected or a condition falls out of range.

Together, these components are designed to reduce crop losses, cut down on reactive pesticide use, and give urban farmers a more consistent, less labor-intensive way to manage their hydroponic systems.

2.4 Target Users

AI-Droponics is built around the needs of two core user groups:

Primary:

- **Urban Farmers.** Farmers operating small-to-mid-scale hydroponic systems, such as those at Gulayan at Bulaklakan, who need continuous environmental monitoring, automated corrective action, and early disease alerts without having to check conditions manually throughout the day.

Secondary:

- **Hydroponics Enthusiasts and Hobbyists.** Individuals setting up smart, automated home or community hydroponic systems who want the same level of monitoring and automation as commercial operations, without the time or technical background to manage it manually.

The end beneficiary of the system is the urban farming operation as a whole, through reduced crop loss, less manual labor, and more consistent, sustainable hydroponic production.

Chapter 3: Project Overview

3.1 Innovation Statement

Source advances urban agriculture through **AI-Droponics**, an integrated system that combines IoT-based hydroponic monitoring, automated farm management, and AI-powered crop disease detection. It addresses a key challenge in hydroponic farming: maintaining stable growing conditions while identifying crop diseases early enough to limit their spread within shared, recirculating water systems.

AI-Droponics brings together two complementary technologies in a single platform. Its IoT monitoring and automation component continuously measures essential environmental parameters, including pH, temperature, humidity, and water levels, and enables automated responses with nutrient dosing. Its AI-powered disease detection component analyzes images of crops to recognize potential signs of disease at an early stage, allowing users to respond before the condition becomes widespread. By combining environmental monitoring, disease detection, and automated intervention, the system supports a more proactive approach to hydroponic farm management, helping users improve productivity, reduce manual workload, optimize resources, and minimize crop losses.

AI-Droponics is primarily designed for urban farmers managing small- to medium-scale hydroponic farms, with secondary users including hydroponics hobbyists and enthusiasts who seek an intelligent and automated way to monitor their growing environment and protect crop health.

3.2 Design Thinking Summary

Empathize

Findings: On-site observation and an interview with David Balilla, agriculturist and urban farm owner at Gulayan at Bulaklakan, showed that hydroponic operations relied almost entirely on manual monitoring, with no automated way to track or respond to changing conditions. Several crops were left unmaintained and rotting, showing that

manual checks could not catch problems in time. Pest prevention followed a calendar-based approach built around historically known pest periods rather than real-time signs of disease. Mr. Balila said he would trust an automated system only after seeing demonstrated accuracy and real on-farm results. Interviews with vegetable vendor Lui Magculang and store owner Rosa Gutierrez further confirmed the need for automated disease detection, alerts, and simple treatment explanations.

Define

Problem Statement**:** Urban hydroponic farmers lack an efficient, real-time way to monitor critical growing conditions and detect crop diseases early, resulting in delayed responses, avoidable crop loss, and an over-reliance on manual labor and reactive pesticide use.

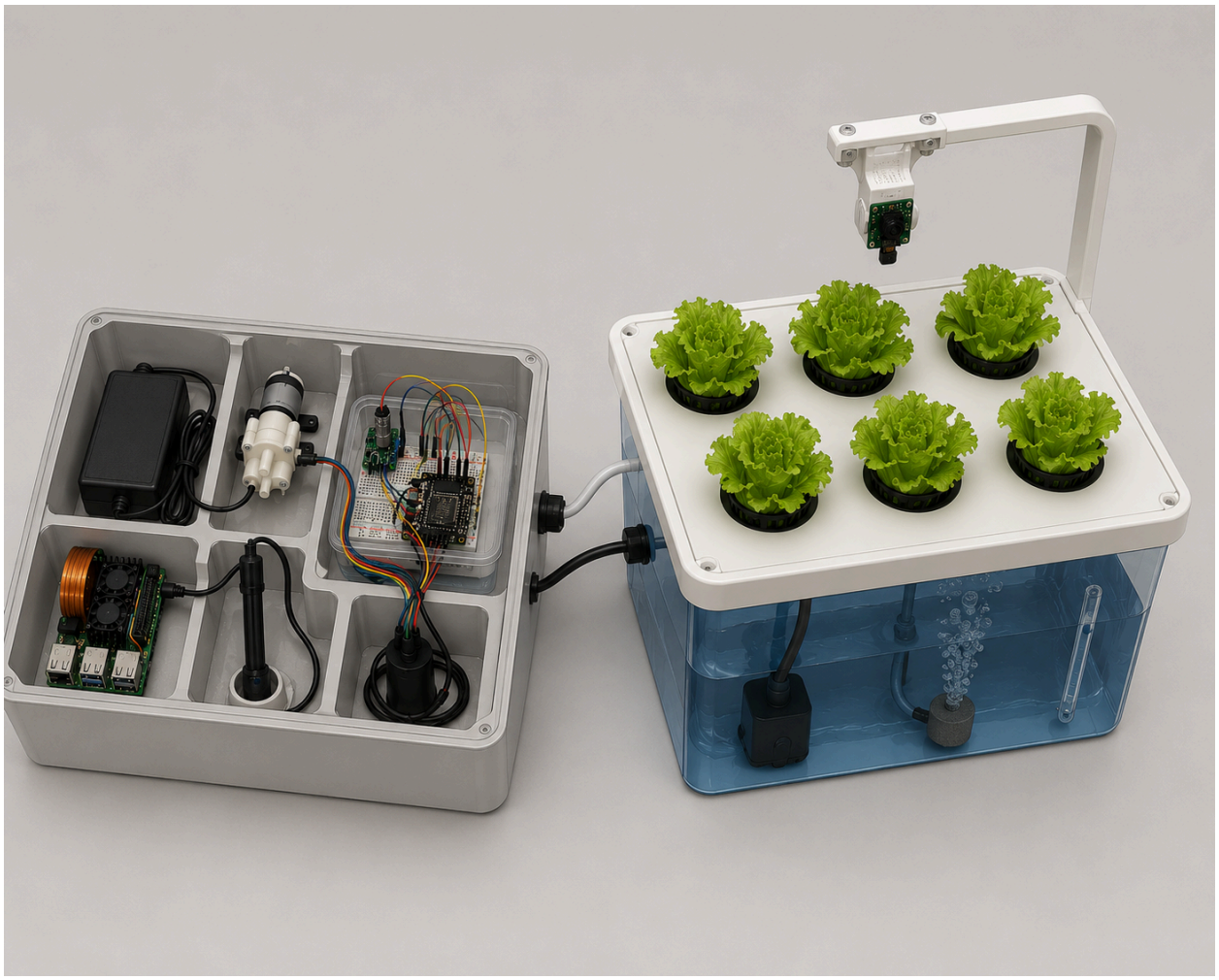
Ideate

The team considered several approaches before selecting the final concept:

- **Basic sensor monitoring system** — tracks environmental conditions only, with no automated response.
- **Cloud-based disease detection system** — relies on server-side AI processing rather than on-device inference.
- **Real-time smart alert system** — limited to notifications, with no automated corrective action.
- **AI advisory app** — recommends treatment, but requires manual application by the farmer.
- **Integrated IoT + AI automation system (selected)** — unifies real-time sensor monitoring, on-device AI disease detection, and automated pump/treatment response within a single mobile app, improving crop protection, farmer response time, and overall system reliability without requiring constant manual intervention.

Prototype

Prototype Description: The team developed an initial prototype of AI-Drophonics consisting of an IoT sensor network and an AI-powered crop disease detection unit. The hardware component utilized a Raspberry Pi 5 with Camera Module 3, pH, DHT, and ultrasonic sensors, along with a relay-controlled water pump and mist nozzle system for automated nutrient dosing. A YOLOv8n-based AI model, run on-device, processes live leaf images to detect early signs of disease. These readings and detections sync to a Flutter-based mobile application through Firebase, giving farmers real-time visibility into environmental conditions, automated corrective actions, and AI-generated disease alerts. Together, these components validated the integration of environmental monitoring, automated response, and on-device AI disease detection within a single platform.



Test

Testing Plan**:** The AI-Drophonics prototype was field-tested at Gulayan at Bulaklakan, with the Raspberry Pi 5 unit, sensors, and AI disease detection model deployed under real hydroponic growing conditions. Testing focused on three areas: how accurately the AI could detect disease, how reliable the sensor readings were, and how responsive the automated pump and treatment system was in practice. Dave Balila, an agriculturist and the project's main client, evaluated both the hardware and monitoring setup as well as the mobile app's usability firsthand on-site. His feedback is now being used to pinpoint areas for improvement, particularly detection accuracy and the clarity of treatment guidance, and will shape further refinements ahead of wider deployment.

3.3 Proposed Innovative Features

Feature	Description	Benefit to Users
Real-Time Environmental Dashboard	Continuously displays pH, temperature, humidity, and water level status readings from connected sensors.	Eliminates manual checking and gives farmers instant visibility into system health
Automated Water pump	Automatically dispenses water pump (refill or drain) based on sensor thresholds	Reduces manual labor and ensures consistent, timely dosing without guesswork
AI-Powered Crop Disease Detection	Analyzes crop images using an AI model to identify early visual signs of disease	Enables early intervention, reducing crop loss and reliance on reactive pesticide use especially critical in water-based systems where disease spreads quickly

Offline-Resilient Automation & Detection	Continues on-device disease detection and automation using stored thresholds during network outages	Ensures crops stay protected and monitored even without a stable internet connection

3.4 Innovation Value

Customer Value

AI-Droponics reduces the manual labor and guesswork urban farmers currently spend on both environmental monitoring and disease inspection, giving them earlier warnings, fewer surprises, and more time to focus on growing rather than troubleshooting.

Social Value

AI-Droponics makes smart hydroponic farming more accessible to urban farmers by reducing the need for constant manual monitoring and providing automated crop management. By supporting efficient resource use and early disease detection, the system promotes more sustainable urban farming and helps communities produce fresh, local food more reliably.

Economic Value

AI-Droponics connects technology with financial benefits by reducing crop losses, minimizing unnecessary use of water, nutrients, and treatments, and lowering the labor required for manual monitoring. These efficiencies can help farmers reduce operating costs, protect their income, and improve the overall profitability of urban hydroponic farming.

Competitive Value

AI-Droponics differentiates itself by combining real-time hydroponic monitoring, AI-powered crop disease detection, and automated corrective actions within a single

platform. Unlike systems that focus only on environmental monitoring, alerts, or standalone disease diagnosis, AI-Droponics connects monitoring, detection, and response, allowing farmers to identify potential problems earlier and take action with less manual intervention. Its hydroponics-specific design and offline-capable AI detection further strengthen its advantage over more general agricultural solutions.

Chapter 4: Startup Opportunity Analysis

Purpose

This chapter assesses whether Source is ready to advance into the Business Development phase. It reviews five key dimensions of the project how well the problem and solution align, the level of market interest, the strength of the innovation, whether the technology can realistically be built and maintained, and whether the team has the capability to execute all in the context of delivering AI-Drophonics, an IoT and AI-based hydroponics monitoring system, to urban farming clients such as Gulayan at Bulaklakan.

4.1 Startup Readiness Assessment

Rating Scale

Rating	Inspiration
5	Excellent / Strongly Agree
4	Good / Agree
3	Fair / Neutral
2	Needs Improvement
1	Poor / Strongly Disagree

Assessment Criteria

Assessment Criteria	Description
Problem–Solution Fit	The proposed startup effectively addresses a real and significant problem experienced by the target users.
Market Demand	There is evidence that potential customers are interested in the proposed

	solution.
Innovation	The startup offers unique features or improvements compared with existing solutions.
Technical Feasibility	The proposed technology, tools, and resources are available and can be implemented by the team.
Team Capability	The team possesses the necessary knowledge, skills, and commitment to complete the project successfully.

Individual Assessment Form

Evaluator: Concepcion, Justine Altaire B.

Assessment Criteria	Rating (1-5)	Remarks / Justification
Problem–Solution Fit	**5**	The AI-Droponics Project is a direct solution to the problem of urban farming in Barangay Holy Spirit. This project recognized the pain points of urban farming here in Quezon City. As the City promotes and encourages urban farming following the SDG for the communities Food Security. Automated hydroponics farm using IOT processes and a system that monitors plant growing thresholds lessening human interaction upon using sensors controlled by ESP32 and Raspberry pi

		5 for AI crop disease detection. AI crop disease will help urban farmers keep track of their crops as they are susceptible to root rot, fungi, bacterial growth and others. Hydroponics are more vulnerable to these diseases and are detrimental to the plant's health, the spread of the diseases is fast and devastating because hydroponics runs on flowing water that circulates throughout the tubing or styrofoam exoskeleton.
Market Demand	**4**	This project has many possible clients such as Department of Science and Technology, Department of Agriculture, Department of Plant Industry, which is also tied up to our selected client which is the Gulayan at Bulaklakan Integrated Natural Farm. The only concern here is we do not have any other urban farm site to data gather if they are experiencing the same problem that AI-Droponics solves.
Innovation	**4**	With integration of IOT, Automation, Machine learning and Artificial Intelligence that implies monitoring, crop disease detection, automatic natural plant treatment solution, and suggestive AI.

Technical Feasibility	**5**	All of the hardware components are achievable to build this project. Software is also bought on subscription and open sources which will be feasible for the project.
Team Capability	**5**	We have great team chemistry, expertise on SDLC, different coding languages and front end to backend implementation, database, mobile app development and applying some industry standards for security and system efficiency. Making sure the planning phase to testing phase to deployment phase will be attainable.

Individual Assessment Form

Evaluator: De Juan, Axel P.

Assessment Criteria	Rating 1-5	Remarks / Justification
Problem–Solution Fit	**5**	We saw the problem firsthand at the farm crops lost to late disease detection and manual monitoring and our solution addresses it directly.
Market Demand	**4**	We've validated the need at our pilot site, but we still need to gather interest from other farms to confirm broader demand.
Innovation	**5**	We're the only ones combining

		automated monitoring with AI disease detection in one system; existing solutions only do one or the other.
Technical Feasibility	**5**	We're confident in the IoT and automation side; we still need to build out and validate our AI disease-detection dataset.
Team Capability	**5**	We have the right mix of skills across our team (IoT, AI, software); we just need to prove it with a working prototype.

Individual Assessment Form

Evaluator: Lingatong, Nazh Manuel V.

Assessment Criteria	Rating 1-5	Remarks / Justification
Problem–Solution Fit	**5**	Source demonstrates a strong understanding of real-world challenges and develops solutions that directly address identified user needs. Its initial product, AI-Droponics, was developed from firsthand observations and customer feedback, establishing a clear connection between the company's innovation approach and the problems it aims to solve.
Market Demand	**4**	Source has established initial market interest through customer interviews,

		feedback from an agriculturist and urban farm owner, and identification of potential customers and institutional partners. However, broader customer validation and additional market research are still needed to confirm demand and support future expansion.
Innovation	**5**	Source demonstrates a strong commitment to innovation by integrating emerging technologies such as AI and IoT into practical solutions. AI-Droponics serves as the company's initial example of combining multiple technologies to address a real-world agricultural challenge, while Source's broader vision allows the company to develop solutions beyond a single product or industry.
Technical Feasibility	**5**	Source possesses the technical foundation required to develop and improve its technology-driven solutions, with team capabilities spanning software development, IoT, AI, and system integration. The company also uses accessible and established technologies, allowing its initial product and future solutions to be developed and refined using available resources.

Team Capability	**5**	Source is a team with complementary skills in technology, software development, AI, IoT, project management, and product development. The team's ability to collaborate, develop a working prototype, gather user feedback, and continuously improve the solution demonstrates its readiness to pursue the company's goal

Individual Assessment Form

Evaluator: Odon, Marc Joseph V.

Assessment Criteria	Rating 1-5	Remarks / Justification
Problem–Solution Fit	**5**	We identified the problem during our farm visit, where late disease detection and manual monitoring caused crop losses. Our solution is designed to address these issues.
Market Demand	**4**	We have validated the need for our solution at our pilot farm, but we still need feedback from more farms to confirm wider demand.
Innovation	**5**	Our system is unique because it combines automated monitoring and AI disease detection into one platform, unlike existing solutions.

Technical Feasibility	**5**	We are confident in our IoT and automation features, but we still need to improve and validate our AI disease detection dataset.
Team Capability	**5**	Our team has the skills in IoT, AI, and software development, and our next step is to build a working prototype.

Individual Assessment Form

Evaluator: Salvador, Gerard Jose N.

Assessment Criteria	Rating 1-5	Remarks / Justification
Problem–Solution Fit	**5**	Urban farmers rely heavily on manual monitoring of crops and environmental factors. Our system automates this, reducing manual labor and human intervention.
Market Demand	**5**	Validated firsthand by a veteran agriculturist, AI-Droponics addresses a real, felt need in urban hydroponic farming and we're actively expanding farmer outreach to confirm demand at scale.
Innovation	**5**	The combination of automation and AI features sets our solution apart and gives confidence it will genuinely help urban farmers.
Technical Feasibility	**5**	Current internet access, technology,

		and tools make these features very achievable.
Team Capability	**5**	Our team has the key skills needed for IoT, AI training, and software development to deliver the project.

Group Summary

After all members have completed their individual assessments, the ratings are summarized below.

Assessment Criteria	**Concepcion, Justine Altaire B.**	**De Juan, Axel P.**	**Lingatong, Nazh Manuel V.**	**Odon, Marc Joseph V.**	**Salvador, Gerard Jose N.**	**Average**
Problem–Solution Fit	5	5	5	5	5	5
Market Demand	4	4	4	4	5	4.2
Innovation	4	5	5	5	5	4.8
Technical Feasibility	5	5	5	5	5	5
Team Capability	5	5	5	5	5	5
Overall Startup Readiness	**4.6**	**4.8**	**4.8**	**4.8**	**5**	**4.8**

Rating Interpretation

Average Score	Interpretation
4.50 - 5.00	Highly Ready
3.50 – 4.49	Ready with Minor Improvements
2.50 – 3.49	Needs Further Development
Below 2.50	Not Yet Ready

Analysis

Overall Readiness Score: 4.80 / 5.00

Interpretation: Highly Ready

Problem–Solution Fit Technical Feasibility, and Team Capability each scored a perfect 5.00, reflecting strong evaluator confidence drawn from firsthand farm visits to Gulayan at Bulaklakan, where manual monitoring and late disease detection were directly observed, and from the team's proven ability to deliver the system using accessible technologies (ESP32, Raspberry Pi 5, Firebase, TensorFlow Lite) and its combined IoT, AI, and full-stack expertise. Innovation scored 4.80, affirming that combining monitoring, automation, and AI disease detection into one platform is a meaningful improvement over existing solutions.

Market Demand was the lowest at 4.20 still within the Highly Ready range, but evaluators noted that validation remains limited to one pilot site and a small respondent pool, despite confirmed interest from the pilot farm and potential institutional partners like DOST, DA, and DPI.

Supported by field visits, direct consultation with client Dave Balila, and the competitor comparison in Section 4.2, the overall score of **4.80** places AI-Droponics in the Highly Ready category, ready to advance to the Business Development phase.

Before the Midterm Startup Validation Defense, the team will:

- Interview additional urban farms to further strengthen market demand validation.
- Continue building and validating the AI disease detection dataset.
- Simplify the app's treatment recommendations into non-technical language.
- Explore flexible pricing options to address affordability concerns.

4.2 Market and Competitor Analysis

This section analyzes the target market for AI-Droponics and identifies existing competitors in the smart agriculture and hydroponics monitoring space.

Target Market

Primary Customers	Urban farmers operating small- to medium-scale hydroponic farms, such as Gulayan at Bulaklakan Integrated National Urban Farm, who need a reliable way to monitor pH, temperature, and humidity while catching crop disease early in a shared, recirculating water system where pathogens can spread quickly.
Secondary Customers	Hydroponics enthusiasts and hobbyists setting up smart, automated home or community growing systems who want the same level of monitoring and automation without needing to check conditions manually throughout the day.

Competitor Analysis

Competitor	Strength	Weaknesses
DOST-PCAARRD / ISAT U Smart Greenhouse Hydroponics https://pcaarrd.dost.gov.ph/index.php/quick-information-dispatch-qid-articles/iot-smart-greenhouse-hydroponics-for-lettuce-makes-its-way-to-farm-enterprise	• Government-validated, hydroponics-specific monitoring (pH, temp, EC, water flow) via Android app.	• No AI disease detection; a single research deployment, not a replicable product.
AgriConnect PH (soil-sensor AI risk app) https://agriconnectph.com/	• AI-driven risk alerts (soil, water, pest) with strong institutional backing and national scale-up plans.	• Soil-based, not hydroponic; alerts only, no automated response; not yet launched.
Plantix plantix.net	• Soil-based, not hydroponic; alerts only, no automated response; not yet launched.	• General-purpose, photo-only diagnosis, no sensor monitoring or automation.

Market Opportunity

Plantix offers strong AI-based disease diagnosis, but depends on the farmer manually scanning each plant with a phone, with no continuous, automated monitoring of the growing environment and no automated response once disease is detected. The DOST-PCAARRD / ISAT U Smart Greenhouse system automates environmental monitoring and dosing for hydroponic setups, but has no AI-based disease detection and remains a single research deployment rather than a replicable product.

AgriConnect PH adds real-time sensor-based alerts, but only for soil-based farms, and still requires the farmer to act manually. None of the three offer continuous environmental monitoring, automated corrective action, and AI disease detection together in a single hydroponics-specific system — the gap AI-Drophonics is built to fill.

4.3 Technology Trends

Trend	Relevance To AI Drophonics
Government-Backed Digital Agriculture	Aligns with national priorities under DA's Digital Agriculture Roadmap and DOST-PCAARRD funding, opening doors to validation and support.
Growth of Edge AI and On-Device Inference	Lets AI-Drophonics run disease detection directly on the Raspberry Pi 5, staying functional without internet access.
Rising Adoption of IoT in Smart Farming	Affordable sensors make AI-Drophonics' real-time monitoring and automation practical even for small farms.
Expansion of Urban and Controlled-Environment Agriculture	Agriculture Growing hydroponic adoption in cities like Quezon City widens the pool of potential users for AI-Drophonics.

Discussion

Taken together, these trends work in AI-Drophonics' favor. Government support for digital agriculture opens the door to funding and partnerships that a small team can tap into, while the maturity of on-device AI models like YOLOv8n makes it realistic to deliver real-time, offline-capable disease detection without the cost of cloud infrastructure directly reinforcing the Technical Feasibility rating discussed in Section

4.1. At the same time, the rise of IoT-enabled farming and the steady growth of urban agriculture point to an expanding base of farmers and hobbyists who are increasingly open to automated monitoring tools. Overall, the timing appears favorable: the technology is mature enough to build on, policy support is in place, and interest from the target market is trending upward.

4.4 Unique Selling Proposition (USP)

USP Statement

AI-Droponics is the only hydroponics-specific system that combines real-time environmental monitoring, AI-powered disease detection, and automated corrective response. It is backed by on-site installation, farmer training, and a 1-year limited warranty. Plantix only diagnoses, the DOST-PCAARRD system only monitors, and AgriConnect PH is soil-based and not yet commercially launched.

At ₱23,000 upfront, AI-Droponics costs more than the free Plantix app. However, Plantix covers diagnosis only and requires the farmer to scan each plant manually. AI-Droponics adds continuous sensors, automated dosing and treatment, on-site setup, and warranty-backed support. Based on the competitor analysis in Section 4.2, the DOST-PCAARRD system and AgriConnect PH are not currently sold to individual farms. AI-Droponics is therefore a ready-to-install, supported option that small urban farms can afford.

Warranty and Service Coverage

Coverage	Terms
Hardware Warranty	1 year from installation. Free repair or replacement of defective components (Raspberry Pi 5, camera, sensors, pump/relay)
Installation Support	Free on-site setup, calibration, and farmer training. Free adjustment of wiring or setup faults within 6 months
Support Response	Remote support within 48 hours. On-site visit within 7 days for covered failures
Exclusions	Misuse, flooding or water damage beyond the housing's rating, power surges, physical damage, and unauthorized modification

Key Competitive Advantages

Advantage	Why it matters
All-in-One Monitoring and Response	Farmers no longer check sensor readings manually and separately decide when to act. AI-Droponics observes conditions and responds automatically, reducing labor and reaction time.
Early Disease Detection	The on-device AI model flags early signs of disease that are easy to miss by eye, which matters in hydroponics where a shared water supply spreads infection quickly.
Works Without a Stable Connection	Detection and automation run locally on

	the Raspberry Pi 5, so crops stay protected even when internet access drops.
Warranty and After-Sales Support	A 1-year warranty, on-site setup and training, and a fast support response give farmers confidence that none of the three competitors currently offer.
Justified Pricing	The ₱23,000 upfront price is higher than free diagnosis-only apps, but it covers hardware, automation, and support. It remains within reach of small farms and hobbyists.

Chapter 5: Customer Validation

5.1 Customer Persona



David Balilla

He/him

- 78 Years Old
- Gulayan At Bulaklakan Research Training Center.
- Agriculturist
- Prefer Not To Say

Background & Lifestyle

An outgoing and charismatic agriculturist, David is known around Gulayan at Bulaklakan for being great with people an extrovert who enjoys engaging with fellow farmers, trainees, and visitors at the research center. Outside the farm, he loves cooking, often turning the produce he grows himself into meals he shares with family and the farming community.

Goals & Motivations

- Adopt tools that make monitoring easier without losing the hands-on control.
- Protect crop yield and quality, both for the farm's output and the fresh produce he loves cooking with
- Continue running his urban farm productively despite the demands of a large growing area

Challenges & Pain Points

- Struggles to manually inspect a large growing area for early disease signsNeeds affordable digital tools for design projects.
- Diseases often go unnoticed until they've already spread through the shared hydroponic water system

Shopping & Buying Behaviour

- Skeptical of marketing claims wants proof through real, on-site demonstrations
- Warms up quickly to new products once trust is earned through visible results his people-oriented nature makes him a strong advocate once convinced

Personality Traits

People Oriented 90%

Experience Driven 90%

Tech-Savvy 40%

Price-Sensitive 50%

Charismatic 80%



Rosa Gutierrez

She/Her

- 75 yrs old
- New Lower Bicutan, Taguig City
- vegetable supplier
- Php 35,000/month
- Widowed

Background & Lifestyle

Rosa is a 75-year-old store owner who grows vegetables at home for her family. She prefers simple, reliable, and affordable tools that make caring for her crops easier.

Goals & Motivations

- To sell a good quality and healthy vegetables
- To avoid the vegetables from poor quality because of diseases
- To have a good feedbacks in regular customers

Challenges & Pain Points

- Figuring the disease in vegetables
- Noticing the vegetables if it does have diseases
- Identifying the diseases of the vegetables

Shopping & Buying Behaviour

- Budget-conscious – carefully considers the cost before buying.
- Checks reviews and feedback from other farmers and gardeners.
- Values affordability, ease of use, and reliability.
- Willing to invest if the system helps prevent crop losses and makes plant care easier.

Personality Traits

Creative 75%

Tech-Savvy 40%

Price-Sensitive 65%

Social Influence 70%

Brand Loyalty 70%



Lui Magculang

He/His

- 21 yrs old
- Riverside San Jose
- Vendor
- Php 1,000/Week
- Criminology
- Single

Background & Lifestyle

A criminology student with a small vegetable store. A farmer of a small farm of vegetables that are being sell in a small place in their home.

Goals & Motivations

- To sell a healthy vegetable
- To prevent the vegetables from sickness and spoilage

Challenges & Pain Points

- Struggle to maintain healthy vegetables to sell, Struggling to pest control

Shopping & Buying Behaviour

- Relies on reviews and feedback from other people before purchasing.
- Trusts products with positive reviews and recommendations.
- Considers user experiences when choosing equipment or products.

Personality Traits

Creative 90%

Tech-Savvy 90%

Price-Sensitive 50%

Social Influence 58%

Brand Loyalty 70%

5.2 Interview Results

Interview Summary Table

Respondent	Role	Key Feedback	Interested (Yes/No)
Lui Magculang	Vegetable Vendor	Needs automated crop disease detection and treatment notifications.	Yes
Rosa Gutierrez	Vegetable and Poultry Store Owner/Supplier at the Local Market	Wants a user-friendly system with AI-powered crop detection, alerts, and simple	Yes

		treatment explanations.	
David Balilla	Agriculturist / Urban Farm Owner	Wants demonstrated accuracy and real on-farm results before trusting the system for his livelihood.	Yes

5.3 Validation Findings

All three respondents expressed genuine interest in AI-Droponics, spanning a wide range of user profiles: a young vegetable vendor, a market store owner, and a veteran professional agriculturist. Across the board, respondents valued the automation of crop monitoring and the promise of early, automated disease detection. Notably, David Balilla's interest as an experienced agriculturist and main client adds strong credibility to the idea, since his endorsement comes from firsthand farming expertise rather than general curiosity.

Common Concerns

Cost and trust were the most recurring concerns. Magculang and Gutierrez both emphasized affordability and wanting proof of value (a trial, demo, or seminar) before committing money. Magculang's buying decisions are driven heavily by other people's reviews, meaning trust has to be earned through visible results and testimonials. Balila, as an expert, raised the bar further; he wanted to demonstrate accuracy and real on-farm results, not marketing claims, before relying on the system for his livelihood. Gutierrez also pointed to gaps outside the app itself, such as expensive or unreliable seeds and rainy-season challenges without a greenhouse.

Overall Assessment

Customer interviews validated the core problem-solution fit of AI-Droponics every respondent confirmed interest, and the need for easier, earlier disease detection came up consistently across very different types of growers. Having David Balila, a 50-to-60-year farming veteran and the project's main client, validate the idea firsthand strengthens this beyond a typical small customer sample. That said, with only three respondents interviewed, this validation remains preliminary; the recurring emphasis on cost, proof of accuracy, and training needs shows that trust and affordability, not the core concept, are the main barriers to adoption that still need to be addressed before scaling.

5.4 Recommended Improvements

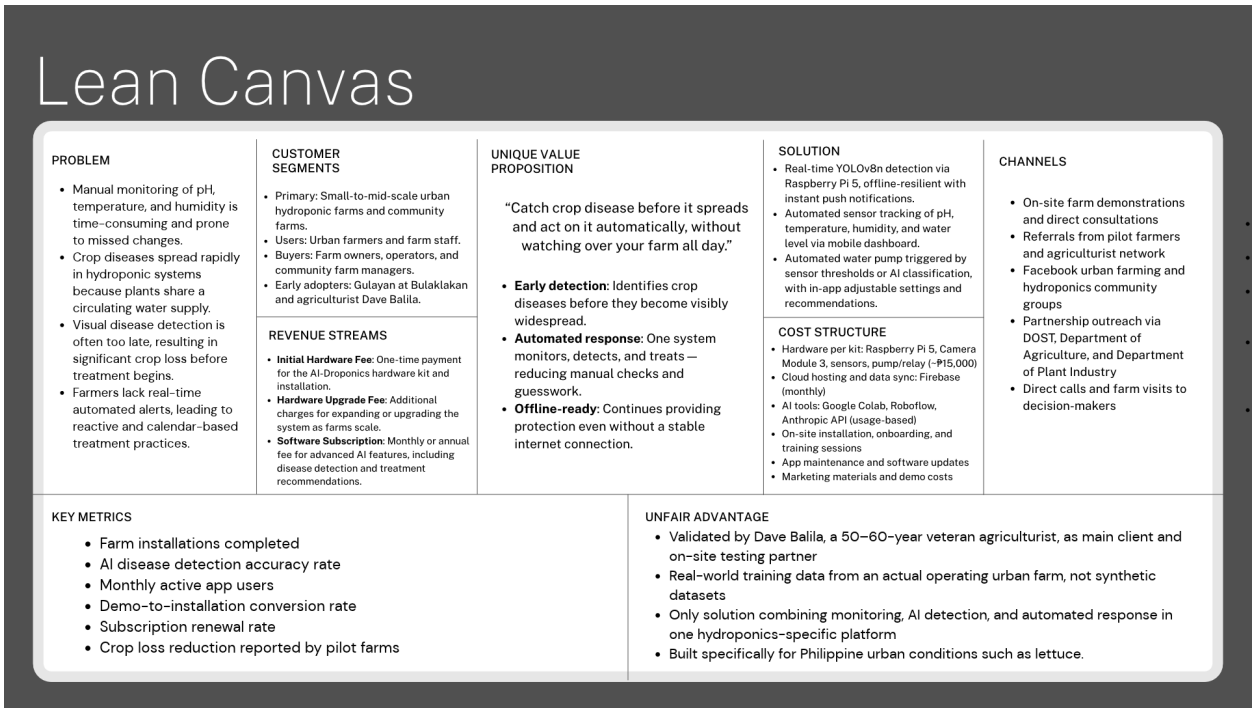
Based on the feedback gathered, the following improvements are recommended:

1. **Offer a demo or trial period** — Gutierrez and Balila indicated they'd trust the system more once they could see it in action rather than just hearing the concept explained; a future demo or trial period, potentially held at Balila's farm, could help build that confidence.
2. **Publish accuracy data and results** — Balilla suggested the system will need to prove its detection accuracy with real results rather than marketing claims once testing begins, transparent accuracy data would also give reviewer-driven buyers like Magculang credible proof to share.
3. **Introduce flexible/affordable pricing options** — Gutierrez raised cost as a concern based on the concept alone offering flexible or tiered pricing could make the system more accessible to budget-conscious hobbyists and small vendors once it's ready.
4. **Provide hands-on training and seminars** — Gutierrez specifically requested hands-on learning opportunities for when the system becomes available

onboarding workshops could help build user confidence and ease the transition to new technology.

5. **Simplify treatment guidance language** — Based on the concept presented, Gutierrez suggested treatment guidance be explained in simpler, non-technical language keeping future AI-generated recommendations plain would make them easier to understand.

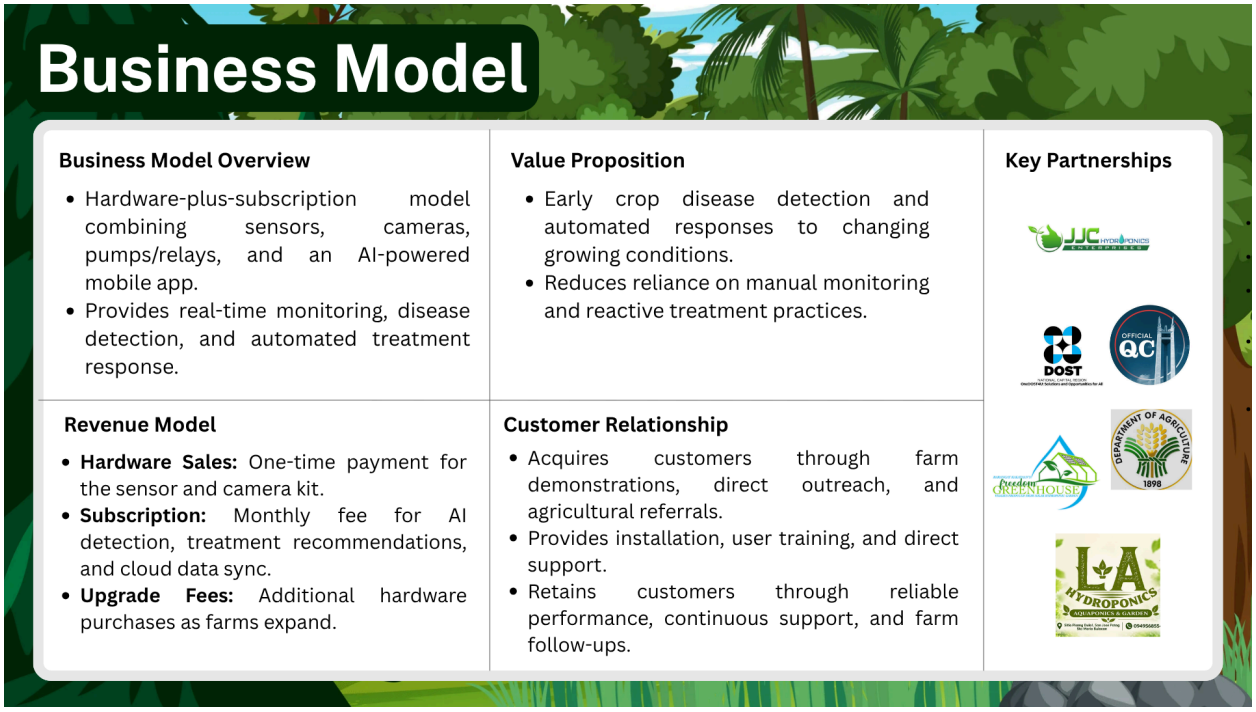
Chapter 6: Lean Canvas



AI-Droponics tackles the slow, error-prone manual monitoring and late disease detection common in small-to-mid-scale urban hydroponic farms, where shared water systems let crop disease spread fast and visible symptoms often appear only after major losses. Validated with agriculturist Dave Balila at Gulayan at Bulaklakan, the system pairs real-time sensor monitoring with YOLOv8n-based AI disease detection on a Raspberry Pi 5 and automated pump control, letting farmers catch and treat problems early without constant supervision, even offline. Revenue comes from a one-time hardware/installation fee, upgrade fees as farms scale, and a subscription for advanced AI features, covering costs for the ₱15,000 hardware kit, cloud hosting, AI tooling, and support. Growth is driven through farm demos, referrals, community outreach, and partnerships with DOST, DA, and DPI, and tracked via installations completed, detection accuracy, active users, and renewal rate. Its edge: real farm-validated data and the

only platform combining monitoring, AI detection, and automated response built specifically for Philippine urban hydroponics.

Chapter 7: Business Model



Chapter 8: Minimum Viable Product (MVP)

8.1 MVP Description

AI-Droponics' MVP is a working smart hydroponics prototype that combines real-time environmental monitoring with AI-powered crop disease detection through a mobile application. It can collect key growing conditions such as pH, temperature, humidity, and water level, while the AI model analyzes captured crop images to identify potential diseases and provide their severity and recommended treatment. The MVP also demonstrates automated responses through water pumping and treatment mechanisms, with the mobile application serving as the main interface for viewing system data, detection results, and alerts.

The MVP is primarily intended for urban farmers and small-scale hydroponic growers, particularly users who experience difficulties with manual monitoring and early identification of crop diseases. These features were prioritized based on customer validation, where users identified manual monitoring, difficulty detecting diseases early, and the need for timely crop protection as important concerns. Feedback also showed that potential users value ease of use, reliable detection, and clear treatment guidance. Therefore, the MVP focuses first on validating the core workflow of monitoring → detecting → intervention, while more advanced features, broader crop coverage, and further commercialization can be developed in later versions.

8.2 Features Included

Feature	Description	Purpose / User Need	Status
Environmental Monitoring	Collects real-time pH, temperature, humidity, and water-level readings through connected sensors.	Reduces the need for frequent manual checking of hydroponic conditions.	Functional

AI Crop Disease Detection	Analyzes captured crop images using the YOLOv8n model to identify potential diseases and their severity.	Helps users identify possible crop diseases earlier and support timely intervention.	Tested
Automated Nutrient Dosing	Activates the nutrient pump based on configured system conditions.	Reduces manual nutrient application and helps maintain appropriate growing conditions.	Functional
Mobile Monitoring Application	Displays sensor readings, disease detection results, alerts, and system status through the Flutter Android application.	Gives farmers a centralized and convenient way to monitor their hydroponic system.	Functional
Disease Alerts & Recommendations	Provides disease information, severity, and recommended treatment steps through the application.	Helps users understand detected crop problems and determine appropriate next steps.	In Development
Crop Image Capture	Captures crop images through the Android device or	Provides the images needed for disease detection without relying	Tested

	Raspberry Pi camera for AI analysis.	solely on visual inspection.	
Data Storage	Stores sensor readings, disease detection events, and system logs in Firebase.	Allows users and the system to retain relevant farm and detection records.	Functional

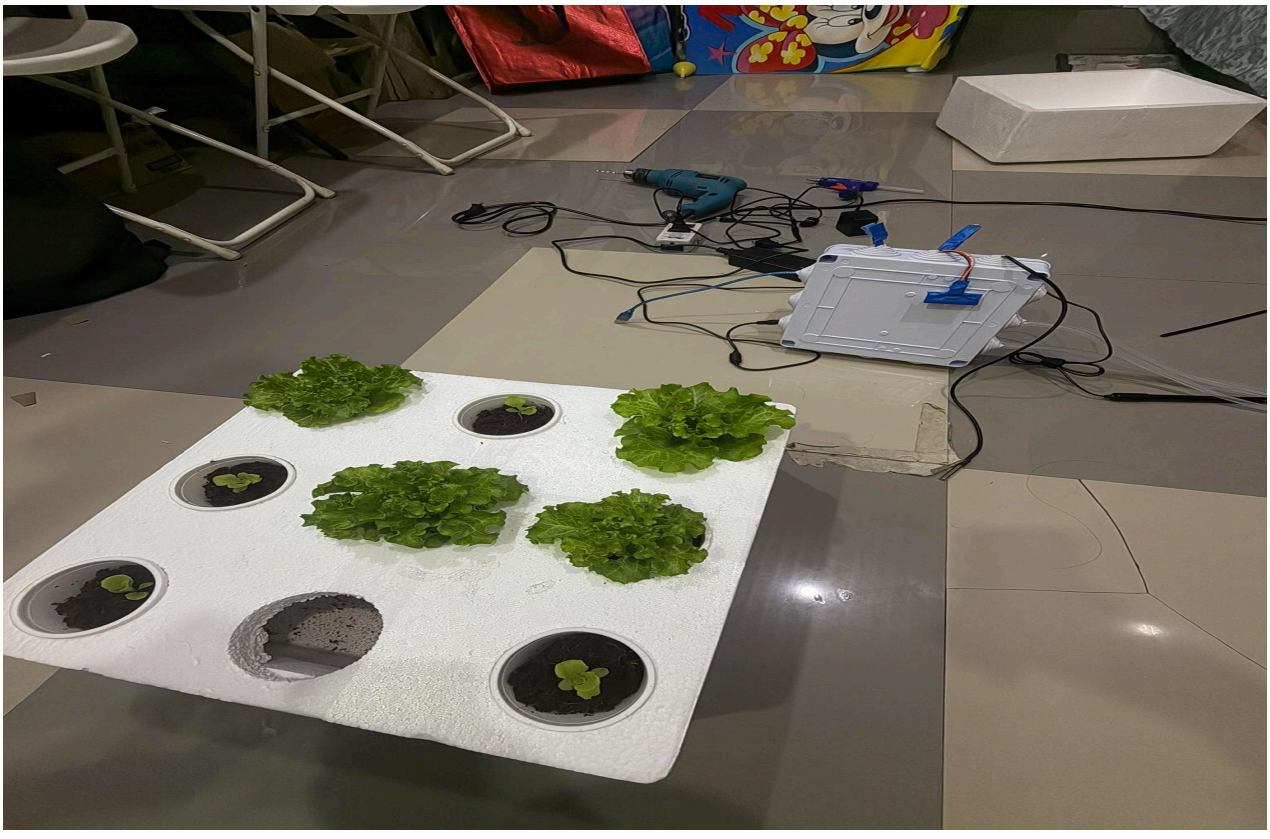
8.3 Product Roadmap

Stage	Target Focus	Plannable Deliverables	Validation Goal
MVP / Version 1.0	Core monitoring and disease detection	Environmental monitoring, AI crop disease detection, crop image capture, mobile monitoring, basic alerts and automated system controls	Confirm that users can monitor hydroponic conditions and identify potential crop diseases through the core workflow
Version 1.1	Usability and reliability	Improve mobile app interface, refine disease detection results, improve alerts and treatment recommendations, strengthen error	Determine whether users can operate the system easily and understand its results and recommendations

		handling and system stability	
Version 2.0	Advanced automation and analytics	Expanded automated treatment controls, historical farm data, trend analysis, improved AI model performance, and broader crop/disease coverage	Test whether advanced automation and analytics provide additional value and improve users' willingness to adopt the system
Future	Expansion and commercialization	Support for additional hydroponic setups, expanded crop coverage, additional integrations, and scalable deployment	Evaluate demand, scalability, and opportunities for wider adoption beyond the initial target users

8.4 Screenshots/Prototype





Chapter 9: Marketing Plan

9.1 Target Market

Target-Market Attribute	AI-Droponics
Primary Customers	Urban farmers and small- to medium-scale hydroponic farm operators
Secondary Customers	Hydroponics hobbyists, community farms, and urban agriculture organizations
Geographic Focus	Initially focused on urban farming operations within Metro Manila and nearby urban areas
End Users	Farmers, farm operators, and individuals responsible for monitoring and maintaining hydroponic systems
Decision-Makers / Buyers	Farm owners, operators, community farm managers, or organizations managing hydroponic facilities
Key Need	Easier monitoring of growing conditions, earlier crop disease detection, and reduced manual intervention
Adoption Indicator	Currently uses or is interested in hydroponic systems and experiences challenges with manual monitoring, crop disease identification, or farm management

Purchase Considerations	Affordability, reliable AI detection, ease of use, demonstrated results, and practical benefits to farm operations

9.2 Market Segmentation

Segmentation Basis	Segment	Meaning for Marketing
Demographic / Firmographic	Small-to-medium urban farms and individual hydroponic growers	Offer affordable and scalable options that fit different farm sizes and budgets.
Geographic	Urban farming operations in Metro Manila and nearby urban areas	Prioritize local demonstrations, pilot testing, and on-site support during initial adoption.
Psychographic	Farmers and growers who value efficiency, sustainability, and technology-assisted farming	Emphasize reduced manual monitoring, efficient resource use, and smart farming capabilities.
Behavioral	Growers who currently rely on manual monitoring and visual crop inspection Technology-curious growers who are interested in automation but require proof before adopting	Demonstrate how AI-Droponics can simplify monitoring and help identify potential crop problems earlier. Provide demonstrations, trial deployments, accuracy results, and testimonials to build trust.

9.3 Pricing Strategy

Pricing Unit: AI-Droponics uses two pricing units: (a) one hardware kit per farm installation (one-time), and (b) one monthly subscription per farm for continued AI detection and cloud features (recurring).

Pricing Input	Amount / Basis	Evidence to Cite
Variable cost per hardware kit	₱15,000 (Raspberry Pi 5, Camera Module 3, sensor set, pump/relay mechanism)	Team assumption based on typical component market prices; to be confirmed by supplier quotation
Monthly variable operating cost	₱8,000 (cloud hosting, Colab/Roboflow, Anthropic API, support allowance)	Team assumption based on current subscription/service rates the software services can still increase in the near future.
Expected monthly sales volume	3 farm installations/month	Team capacity assumption (based on the on-site demo/install process each sale requires) anchored to that narrow, localized market, not a market-research number
Allocated variable cost per unit	₱8,000 ÷ 3 = ₱2,667	Cost-plus computation
Full cost per unit	₱15,000 + ₱2,667 = ₱17,667	Cost-plus computation
Preliminary price	₱17,667 + ₱5,300 = ₱22,967 (≈ ₱23,000)	Cost-plus computation
Final proposed price	₱23,000 one-time hardware fee;	Pricing reflects affordability-adjusted

		willingness-to-pay findings from stakeholder interviews. As installation volume scales, the monthly subscription is designed to become the primary revenue driver, sustaining ongoing investment in the system's AI capabilities.

9.4 Promotion Strategy

Channel / Activity	Purpose	Why it Fits the Target Market	Example KPI
On-Site Farm Demonstrations	Generate qualified leads	Farm owners/operators need to see the monitoring and detection workflow in action before trusting it with their crops.	Demo-to-installation conversion rate
Facebook Content	Build awareness and credibility	Urban farmers and hydroponics hobbyists are reachable through local farming/gardening community groups	Qualified inquiries
Referral Program	Encourage word-of-mouth	Farm operators trust peer	Referral leads

		recommendations, as shown by Magculang's review-driven buying behavior	
Agricultural Agency Partnerships (DOST, Department of Agriculture, Department of Plant Industry)	Reach decision-makers through trusted institutions	Positions AI-Droponics as credible and government-recognized to a cautious, adoption-sensitive market	Partnership-driven leads
Direct Outreach (calls/visits)	Schedule demos/meetings	Supports direct contact with farm owners and operators, the actual purchase decision-makers	Reply/demo-booking rate

AI-Droponics will prioritize trust-building, in-person channels over broad advertising, since Chapter 5 validation showed farmers are cautious adopters who want proof before committing. On-site demonstrations and referrals carry the most weight for driving qualified leads to trial, while Facebook content and agency partnerships build awareness and credibility ahead of the direct sales conversation.

9.5 Distribution Channels

Channel	Purpose	Access/Delivery Method
Direct Sales / On-Site Consultation	Purchase decision and agreement	Farm owner/operator agrees to purchase after an on-site

		demo or consultation
On-Site Hardware Installation	Product delivery	Technical team installs the sensor, camera, and pump/relay kit directly at the farm
Mobile Application (Flutter + Firebase)	Ongoing access	Farmer receives login access to the app for real-time monitoring, alerts, and treatment recommendations after installation
Direct Support Channel	Onboarding and troubleshooting	Farmers can reach the team directly (call/chat) for setup help and ongoing support

AI-Droponics is distributed as a hardware-plus-app service rather than a purely digital product. After a farm agrees to purchase, the team installs the physical sensor and camera kit on-site, configures the system for that farm's specific setup, and creates the farmer's account on the mobile app. Once installation is complete, the farmer accesses real-time monitoring, disease alerts, and treatment recommendations directly through the app, while ongoing support and subscription renewal are handled through direct contact with the team.

Chapter 10: Operations Plan

10.1 Organizational Structure

10.2 Team Roles

Team Member	Role	Responsibility	Expected Deliverables
Concepcion, Justine Altaire B.	Hardware/IoT Lead	Sensor integration, Raspberry Pi 5 setup, pump/relay system assembly and wiring	Functional field station hardware, hardware test logs
De Juan, Axel P.	Documentation Lead	Compiles and maintains all project documentation across Chapters 1-11, including the SRS, test plans, and technical reports. Ensures consistent formatting, numbering, and IEEE citations, and checks that prices, costs, and customer data match across chapters. Tracks	Finalized SRS and test documentation; complete, formatted Final Startup Portfolio (Chapters 1-11); updated Table of Contents, figure/table numbering, and IEEE reference list; labeled appendices; revision-controlled records and feedback log

		adviser and instructor feedback and organizes the appendices.	
Lingatong, Nazh Manuel V.	Project Manager	Coordinates the team's schedule, tasks, and milestones. Develops the Flutter mobile app and Firebase integration.	Project schedule and task tracking, working Flutter mobile app, Firebase integration
Odon, Marc Joseph	QA Tester	Tests system features and reliability. Manages cloud data storage, sensor data pipelines, and system logs.	Test cases and test results, bug reports, working data pipeline, system logs
Salvador, Gerard Jose N.	AI/ML Lead	Trains and validates the YOLOv8n disease detection model. Handles the business plan, customer validation, pricing, and partnerships.	Trained and validated AI model, business plan, customer interview records, pricing computation, partnership outreach

10.3 Development Workflow

Validated Need	Plan	Build	Test	Improve
Farm visits and customer interviews	Prioritize features and assign tasks	Develop hardware, AI model, and app features	Field test at Gulayan at Bulaklakan	Retrain the model, fix issues, update the backlog

The team follows an iterative workflow. Needs validated through farm visits and interviews are turned into prioritized tasks, which the team builds across hardware, AI, and the mobile app. Each build is tested on the pilot farm, and feedback from Dave Balila and the farm staff is documented and returned to the backlog so the next cycle addresses the most important issues first.

10.4 Resources Required

Resource Category	Specific Resource	Purpose	Availability / Source
Purpose	5-person founding team (hardware, AI/ML, app dev, data/backend, business/ops)	Build, test, and maintain the hardware and app	Internal team
Hardware	Raspberry Pi 5, Camera Module 3, pH/temperature/humidity/water-level sensors, pump/relay	Core components for each farm installation	Purchased per unit; team-sourced

	mechanism		
Hardware (Development)	Development laptops, spare/test sensor units	Build and test the MVP before field deployment	Team-owned / shared
Software / AI Tools	Google Colab, Roboflow	AI model training and datasets management	Free/student plans where applicable
Software / AI Tools	Anthropic AI (API access)	AI-generated treatment recommendations	Paid, usage-based
Cloud / Internet	Firebase (hosting, database, data sync)	Operate the mobile app backend and store sensor/detection data	Paid monthly
Validation / Field Access	Pilot site access (Gulayan at Bulaklakan), agriculturist input (Dave Balila)	Real-world testing, model validation, and feedback	Partner-provided
Budget	Marketing and pilot support fund	On-site demos, promotional materials, training/onboarding sessions	Team/project budget

10.5 Risk Management

Risk	Likelihood	Impact	Mitigation
AI detection accuracy below farmer expectations	Medium	High	Continued on-farm testing and model retraining with Roboflow datasets before wider rollout
Farmer reluctance to adopt due to cost	Medium	High	Tiered/flexible pricing, optional training, pilot demonstrations
Hardware component failure in field conditions	Medium	Medium	Use durable, weather-appropriat e housing; maintain spare component stock
Connectivity issues at farm sites	Medium	Medium	On-device (offline-capable) detection reduces dependency on stable internet
Slow B2B sales cycle	High	Medium	Focus on referrals and demos rather than paid advertising to control costs

Chapter 11: Financial Plan

11.1 Financial Assumptions and Evidence

AI-Drophonics uses a hardware-plus-subscription revenue model. The hardware kit selling price of ₱23,000 was derived through cost-plus computation in Section 9.3 and validated against customer willingness-to-pay signals gathered during interviews. The monthly subscription price of ₱500 reflects affordability feedback from respondents Magculang and Gutierrez, who flagged cost sensitivity as a key concern.

The hardware variable cost of ₱15,000 per kit is based on the team's bill of materials covering the Raspberry Pi 5, Camera Module 3, pH/temperature/humidity/water-level sensors, and pump/relay mechanism. Fixed operating costs of ₱8,000 per month cover Firebase cloud hosting, Google Colab and Roboflow subscriptions, Anthropic API usage, and app maintenance allowance. The expected monthly sales volume of 3 hardware kit installations is a conservative team assumption anchored to the on-site installation capacity required per sale. All assumptions are labeled below.

Financial Assumption	Value	Basis / Evidence
Hardware kit selling price	₱23,000/kit	Section 9.3 cost-plus computation + interview willingness-to-pay check
Monthly subscription price	₱500/month per farm	Affordability feedback from customer interviews (Magculang, Gutierrez)
Hardware variable cost per kit	₱15,000	Team bill of materials: Raspberry Pi 5, Camera Module 3, sensors, pump/relay mechanism.
Monthly fixed operating cost	₱8,000	Firebase, Colab/Roboflow, Anthropic API, app maintenance — team assumption based on current subscription/service rates
Expected Month 1 hardware sales	2 kits	Conservative estimate based on pilot site interest and on-site installation capacity
Expected sales growth	+1 kit/month (Months 2–4); stable at 3–4 kits/month thereafter	Team assumption based on planned outreach and demo capacity
Existing subscription base growth	+2 farms/month	Tied to hardware installation rate; each installation adds one subscriber

11.2 Startup Cost

Startup cost covers all one-time pre-launch cash requirements needed before the first paying customer is served. The largest cost item is the initial hardware inventory for the pilot deployment, followed by AI model development tools and business registration. Development laptops and personal devices are team-owned and contributed resources, not counted as new cash expenses.

One-Time Cost Item	Quantity	Unit Cost	Total	Basis/Source
Initial hardware kit inventory (pilot units)	2 kits	₱ 15,000	₱ 30,000	Team bill of materials based on PROJMAN documentation.
Raspberry Pi 5 + Camera Module 3 (dev/test unit)	1 set	₱6,500	₱6,500	Team assumption based on local market prices
Roboflow dataset annotation subscription (initial setup)	1	₱3,000	₱3,000	Team assumption based on Roboflow pricing
Google Colab Pro (initial AI model training)	1	₱1,500	₱1,500	Team assumption based on current Colab rates
Firebase initial setup and first-month hosting	1	₱1,000	₱1,000	Team assumption based on Firebase Blaze plan rates
Business registration and documentation	1	₱3,000	₱3,000	DTI registration estimate team assumption

Launch and demo materials (printed, farm visit expenses)	1 set	₱2,500	₱2,500	Team assumption
Total Startup Cost	**₱47,500**			

Interpretation : ₱47,500 is the estimated one-time cash needed before regular operations begin. The largest single driver is the initial hardware inventory for two pilot kits (₱30,000), which is required to demonstrate the system to prospective farm clients. AI development tools and registration account for the remaining ₱7,500. Development devices used by the team are contributed resources and are not included in the cash requirement.

11.3 Revenue Streams

AI-Drophonics earns income from two streams: a one-time hardware kit sale per farm installation and a recurring monthly subscription per farm for continued AI detection and cloud features.

Revenue Stream	Price/Fee	Expected Customers	Frequency	Estimated Value
Hardware Kit Sale	₱23,000/kit	2 farms (Month 1)	One-time per installation	₱46,000
Monthly Subscription	₱500/farm	2 farms (Month 1)	Monthly recurring	₱1,000
Estimated First-Month Revenue	₱47,000			

11.4 Cost Structure

Fixed Monthly Operating Costs (costs that do not change with sales volume):

Fixed Cost Item	Monthly Amount	Basis / Source
Firebase cloud hosting and data sync	₱2,500	Team assumption based on Firebase Blaze plan rates
Google Colab Pro (ongoing model retraining)	₱1,500	Team assumption based on current Colab rates
Roboflow dataset management subscription	₱1,500	Team assumption based on Roboflow pricing
Anthropic API (AI treatment recommendations)	₱1,500	Team assumption based on usage-based pricing
App maintenance and software updates allowance	₱500	Team assumption
Marketing, demo transport, and outreach activities	₱500	Team assumption
Total Monthly Fixed Costs	₱8,000	

Variable Cost per Hardware Kit (costs that increase with each unit sold):

Variable Cost Item	Monthly Amount	Basis / Source
Raspberry Pi 5	₱5,500	Team assumption based on local market prices
Camera Module 3	₱1,500	Team assumption based on local market prices
pH sensor, DHT sensor, ultrasonic sensor set	₱2,500	Team assumption based on local component market prices
Relay module + water	₱3,000	Team assumption based

pump + mist nozzle mechanism		on local component market prices
Enclosure, wiring, assembly materials	₱1,500	Team assumption
Installation transport and materials	₱1,000	Team assumption
Total Variable Cost per Kit	₱15,000	

Sample Computation at 3 kits sold:

- Total Variable Cost = 3 × ₱15,000 = ₱45,000
- Total Monthly Operating Cost = ₱8,000 + ₱45,000 = ₱53,000

Interpretation: The hardware component cost (₱15,000/kit) is the dominant cost driver and scales directly with every new farm installation. Fixed costs of ₱8,000/month are primarily cloud and AI tool subscriptions that remain constant regardless of sales volume. As the subscription base grows, the fixed costs become increasingly offset by recurring subscription revenue, gradually reducing the dependence on hardware sales to cover operating expenses. The most significant risk is a month with zero hardware sales, where the ₱8,000 fixed cost would need to be covered by the subscription base alone which only becomes feasible once the farm count reaches 16 active subscribers.

11.5 Sales Forecast

Assumptions: Hardware selling price = ₱23,000/kit; Subscription price = ₱500/farm/month; Variable cost = ₱15,000/kit; Fixed costs = ₱8,000/month.

Month	New	Total	Hardw	Subscri	Total	Total	Fixed	Operati

	Kits sold	Subscribers	are Revenue	ption Revenue	Revenue	Variable Cost	Cost	ng Profit/(Loss)
1	2	2	₱46,000	₱1,000	₱47,000	₱30,000	₱8,000	₱9,000
2	2	4	₱46,000	₱2,000	₱48,000	₱30,000	₱8,000	₱10,000
3	3	7	₱69,000	₱3,500	₱72,500	₱45,000	₱8,000	₱19,000
4	3	10	₱69,000	₱5,000	₱74,000	₱45,000	₱8,000	₱21,000
5	3	13	₱69,000	₱6,500	₱75,500	₱45,000	₱8,000	₱22,500
6	4	17	₱92,000	₱8,500	₱100,5000	₱60,000	₱8,000	₱32,500

Interpretation: The forecast shows positive operating profit from Month 1 because each hardware kit generates a contribution margin of ₱8,000 (₱23,000 − ₱15,000), and selling just two kits covers the ₱8,000 monthly fixed cost with the subscription base adding incremental income. The subscription revenue grows steadily each month as the installed farm base accumulates, eventually providing a growing cushion against months with lower hardware sales. The most uncertain assumption is the monthly kit sales volume; the 2–4 kits/month projection is based entirely on team capacity and pilot site demand rather than confirmed purchase orders, so real performance may vary. Profitability is sensitive to whether new farm clients can be acquired consistently through on-site demonstrations and referrals.

11.6 Break-even Analysis

Using hardware kit sale as the primary pricing unit:

- Selling Price per Kit = ₱23,000
- Variable Cost per Kit = ₱15,000
- Monthly Fixed Costs = ₱8,000

Step 1: Contribution Margin per Kit:

₱23,000 − ₱15,000 = ₱8,000 per kit

Step 2: Break-even Quantity:

₱8,000 ÷ ₱8,000 = 1 kit per month

Step 3: Break-even Sales Revenue:

1 × ₱23,000 = ₱23,000 per month

Interpretation: AI-Drophonics reaches operating break-even by selling just 1 hardware kit per month, which is a notably low threshold given that the team's conservative forecast starts at 2 kits in Month 1. This means the business is resilient to slow months even if a single farm installation covers all fixed operating costs for that month. The low break-even quantity results from the high contribution margin per kit (₱8,000), which is a direct consequence of the hardware selling price being significantly above the variable cost. As the subscription base grows, break-even becomes even easier to maintain since recurring subscription income offsets a portion of fixed costs independently of hardware sales. The most realistic risk is a prolonged period with zero new installations, but the growing subscription base provides an increasing buffer against that scenario over time.

Optional Startup Cost Recovery Estimate:

- Startup Cost = ₱47,500
- Expected average monthly operating profit (Months 1–6) ≈ ₱19,000

- Simple Payback = ₱47,500 ÷ ₱19,000 ≈ 2.5 months

11.7 Cash Flow Projection

Assumptions: Beginning cash = ₱47,500 (startup capital); hardware kits purchased at ₱15,000 variable cost per unit before sale; Firebase and tool subscriptions paid monthly.

Month	Beginning Cash	Cash Inflows (Revenue)	Cash Outflows (Variable + Fixed)	Net Cash Flow	Ending Cash
1	₱47,500	₱47,000	₱38,000	₱9,000	₱56,500
2	₱56,500	₱48,000	₱38,000	₱10,000	₱66,500
3	₱66,500	₱72,500	₱53,000	₱19,500	₱86,000
4	₱86,500	₱74,000	₱53,000	₱21,000	₱107,000
5	₱107,000	₱75,500	₱53,000	₱22,500	₱129,000
6	₱129,500	₱100,500	₱68,000	₱32,500	₱162,000

Interpretation: Cash flow is positive from Month 1 because hardware kit revenue is collected at the point of installation, and the startup capital covers the initial inventory and operating costs. The ending cash balance grows consistently across all six months, indicating that the business does not face a cash shortage under the expected scenario. Month 6 sees the highest cash generation due to the combined increase in hardware sales and a growing subscription base. The main cash flow risk is a delay in client acquisition if farm installations are postponed, hardware revenue would be delayed while fixed operating costs (₱8,000/month) continue, drawing down the cash balance.

Maintaining the startup capital as a buffer ensures the team can sustain at least five months of zero-revenue operations before cash runs out.

11.8 Scenario Analysis

Assumptions held constant: Hardware selling price = ₱23,000/kit; Variable cost = ₱15,000/kit; Monthly fixed cost = ₱8,000. Subscription revenue is included in Total Revenue for each scenario based on cumulative subscriber count.

Scenario	New Kits Sold (Month)	Total Subscribers	Hardware Revenue	Subscription Revenue	Total Revenue	Contribution Margin	Fixed Cost	Operating Profit/(Loss)
Conservative	1	3	₱23,000	₱1,500	₱24,500	₱8,000	₱8,000	₱1,500
Expected	3	7	₱69,000	₱3,500	₱72,500	₱24,000	₱8,000	₱19,500
Optimistic	5	12	₱115,000	₱6,000	₱121,000	₱40,000	₱8,000	₱53,000

Interpretation: Even in the Conservative scenario, AI-Drophonics generates a small positive operating profit (₱1,500) because selling just one hardware kit already covers the full fixed cost, with subscription income adding to the margin. This confirms that the business model is resilient even at very low sales volumes. The Expected scenario

produces a strong ₱19,500 monthly profit, while the Optimistic scenario at five installations nearly triples that figure. The assumption with the greatest effect on financial viability is the monthly hardware kit sales volume a drop from three to one kit significantly reduces profit but does not push the business into a loss, which is a meaningful indicator of the model's durability. The team should prioritize maintaining a consistent pipeline of at least two farm installations per month to stay comfortably above break-even.

References: